

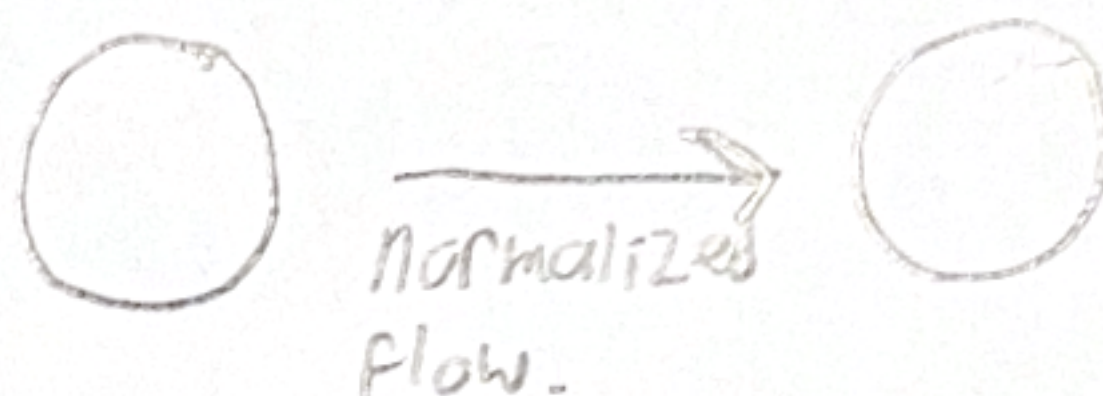
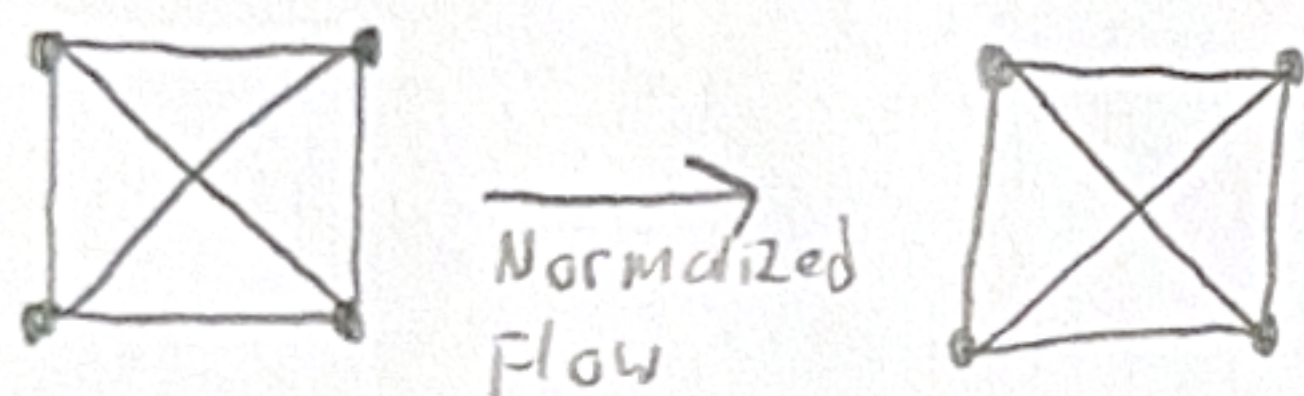
Deleting an edge

There are two different types of edge deletion:

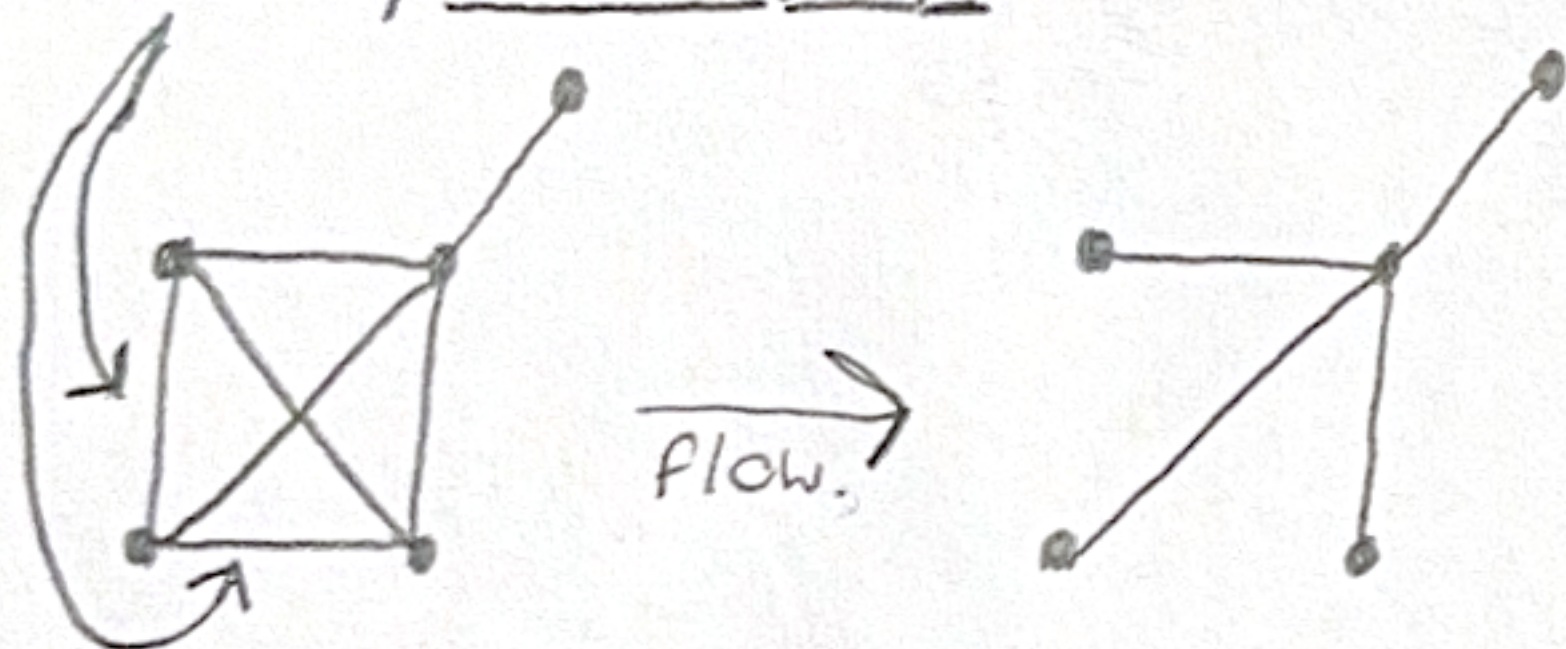
(1) large edge weights that emerge from Ricci flow are analogous to singularities in classical Ricci flow, surgery can be performed.

(2) edge weights can shrink to 0, in which case the edge vanishes and is deleted, which is not a surgery procedure.

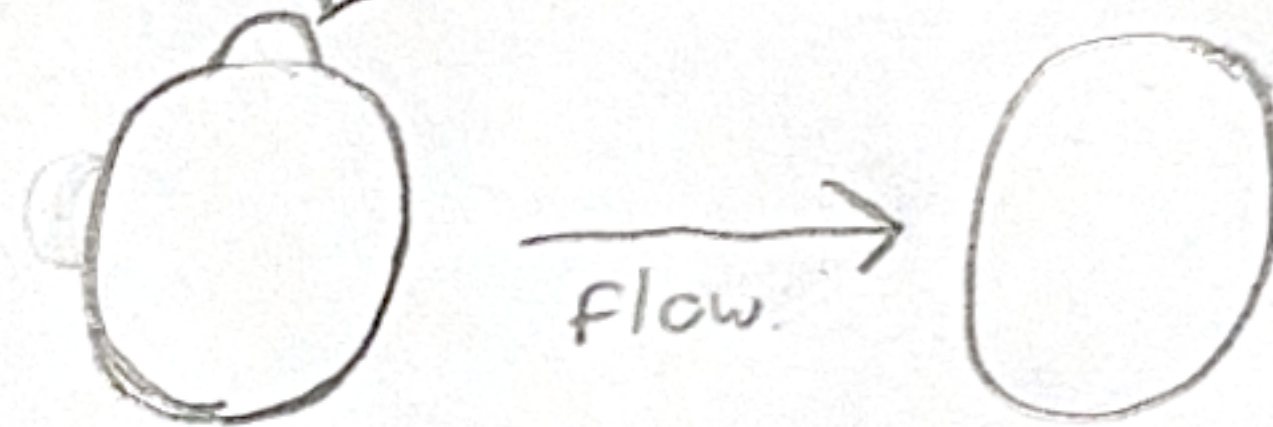
Geometrical meaning / analogy with discrete Ricci flow on graphs?



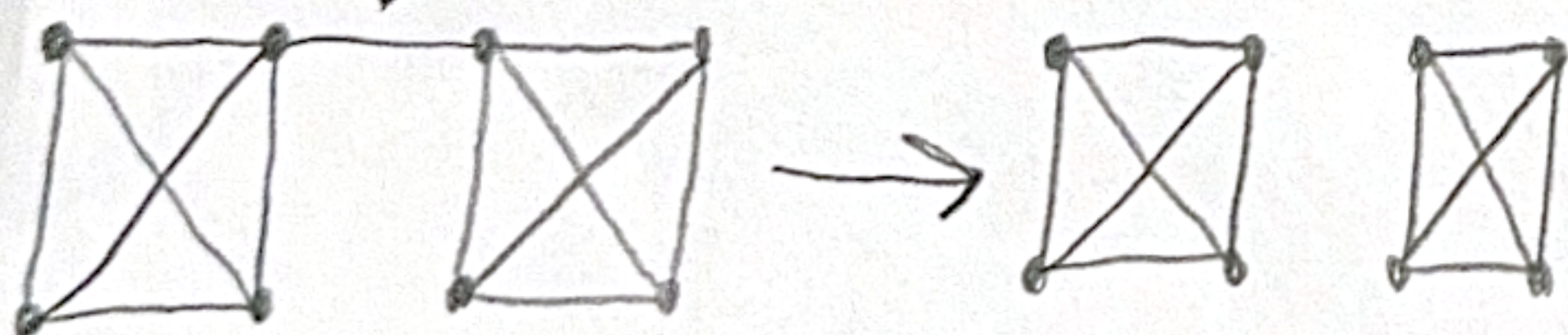
Weight shrinks to 0,
vanishes, no surgery.



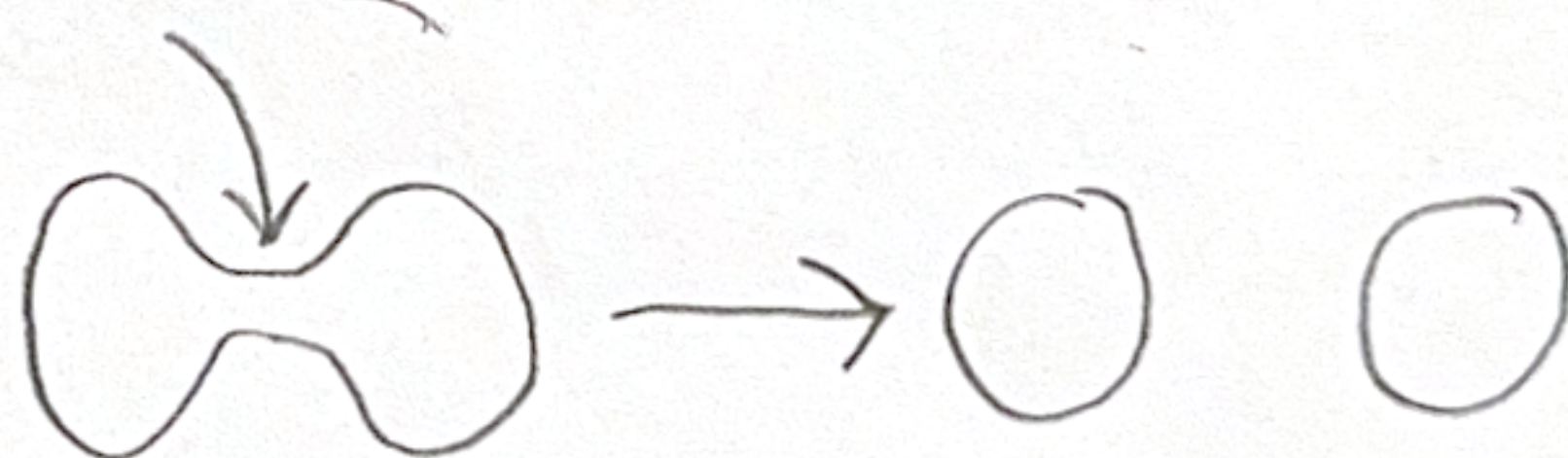
This smooths out and "goes away",
No singularity, no surgery.



Weight grows large,
Surgery performed.



Singularity forms, Surgery preferred.



The important thing is that edge weights approach 0 only in the limit, not finite time, so "continuing" Ricci flow after removing 0 weight edges might not make sense.